

I. New Figures

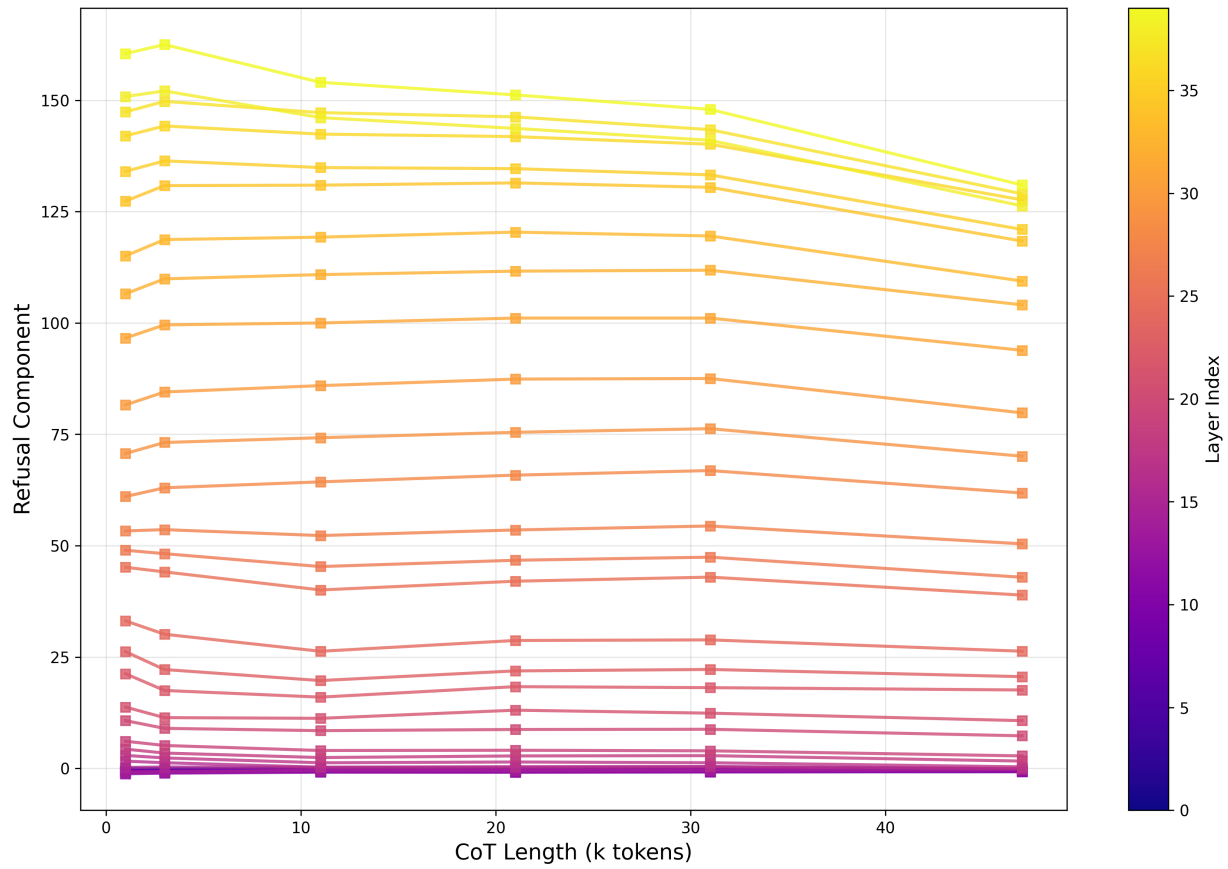


Figure 19. **Refusal component comparison across layers (Qwen3-14B) for different CoT lengths.** Only for harmful instructions. CoT lengths: 1k, 3k, 11k, 21k, 31k, 47k tokens.

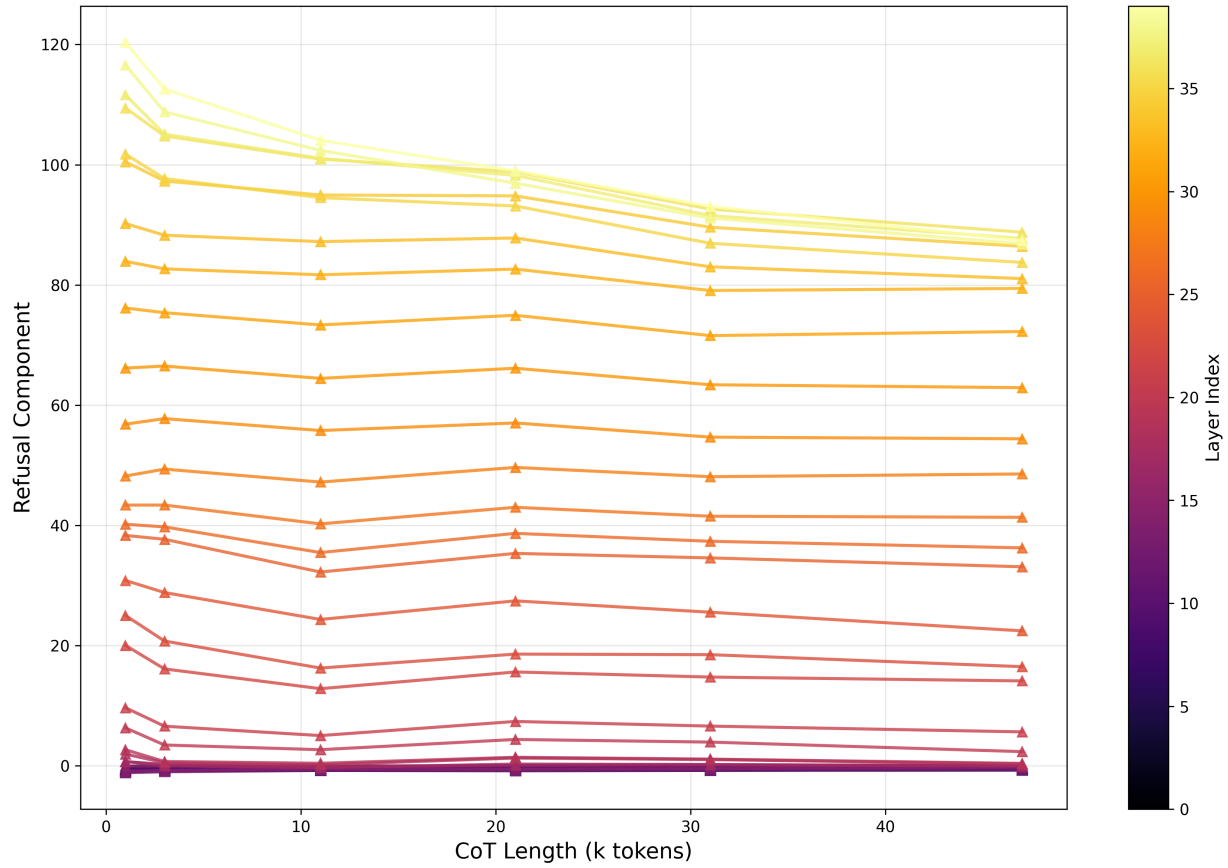


Figure 20. **Refusal component comparison across layers (Qwen3-14B) for different CoT lengths.** Only for stealthy harmful instructions. CoT lengths: 1k, 3k, 11k, 21k, 31k, 47k tokens.

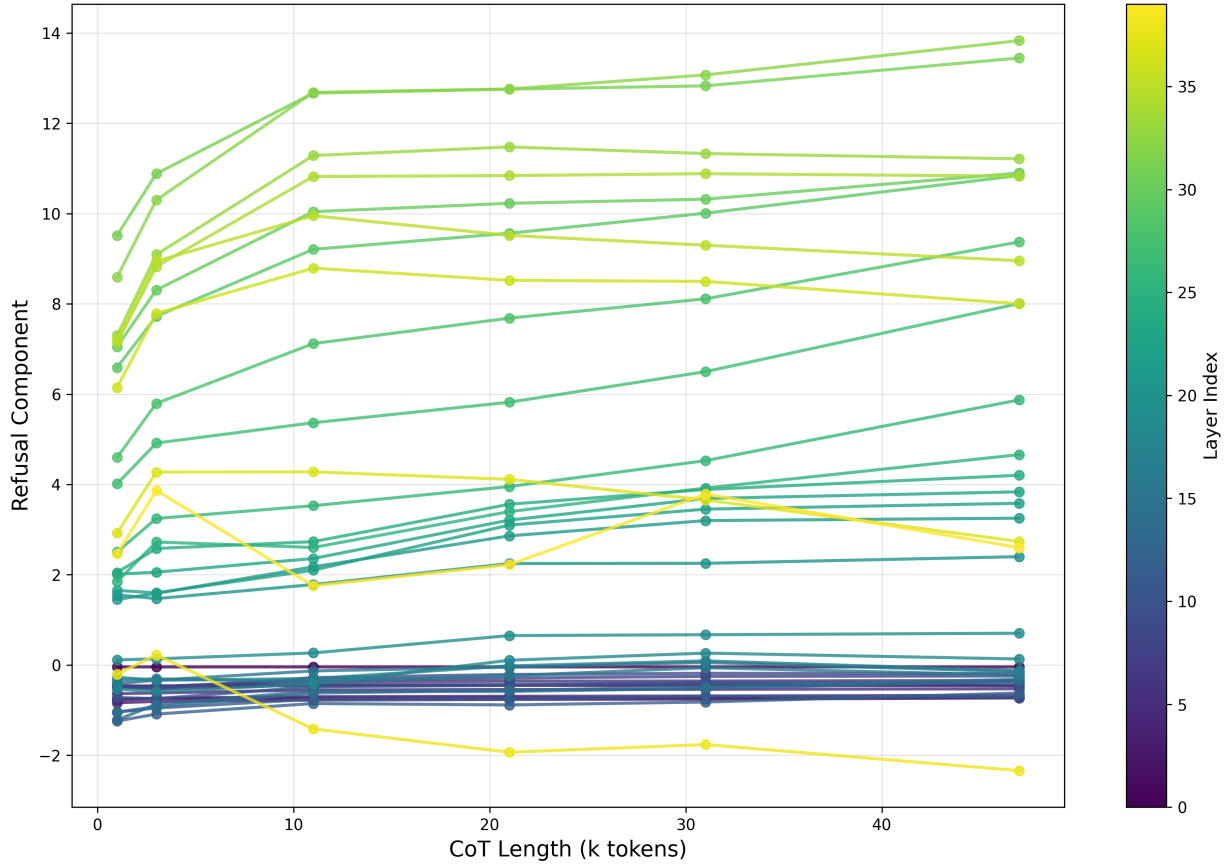


Figure 21. **Refusal component comparison across layers (Qwen3-14B) for different CoT lengths.** Only for harmless instructions. CoT lengths: 1k, 3k, 11k, 21k, 31k, 47k tokens.

J. Attention Visualization (Method + Examples)

Methodology

We analyze attention maps in Qwen3-14B on our jailbreak samples (thinking mode enabled), focusing on how the **first harmful-response tokens** (“target field”) attend over Prompt, Thinking, and Response regions. As shown in Figures 22 and 23, when the model begins generating the first harmful-response token, most attention is paid on the benign reasoning context, with little attention allocated to the ‘harmful instruction’—thereby achieving ‘*refusal dilution*’.